

Studio Ghibli App

Exploring and discovering Studio Ghibli films

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App: [Live App Link](#)
API: [Live API Link](#)



Problem: Content overload, hard to find movies that fit interests - not just genres

Key Objective: Create an app and model for users to explore Studio Ghibli films from images and get recommendations based on their specific interests



Studio Ghibli API
(species,
description)

**The Movie
Database API**
(rating)

Manual Labels
(i.e. 'cats', 'planes',
'forest')

**Official Landscape
Images**
(from Studio Ghibli
website)

Cleaning/Preprocessing: merge on title, many NAs so just keep species, find separate image for 1 movie

Total Movies: 22

Average Features/Movie: 7.27, no movies with 0 features

title	genres	tmdb_rating	description	species	labels	image_path
My Neighbor Totoro	['Fantasy', 'Animation', 'Family']	8.1	Two sisters move to the country with their father in order to be closer to their hospitalized mother, and discover the surrounding trees are inhabited by Totoros, magical spirits of the forest. When the youngest runs away from home, the older sister seeks help from the spirits to find her.	['Human', 'Totoro', 'Cat']	['cats', 'family', 'forests', 'friendship', 'slice_of_life']	C:\Users\mieme\PycharmProjects\STATS418_Project\data\movie_images\My_Neighbor_Totoro\image.jpg

Content-Based Recommender

Input: Specific genres, labels, character species

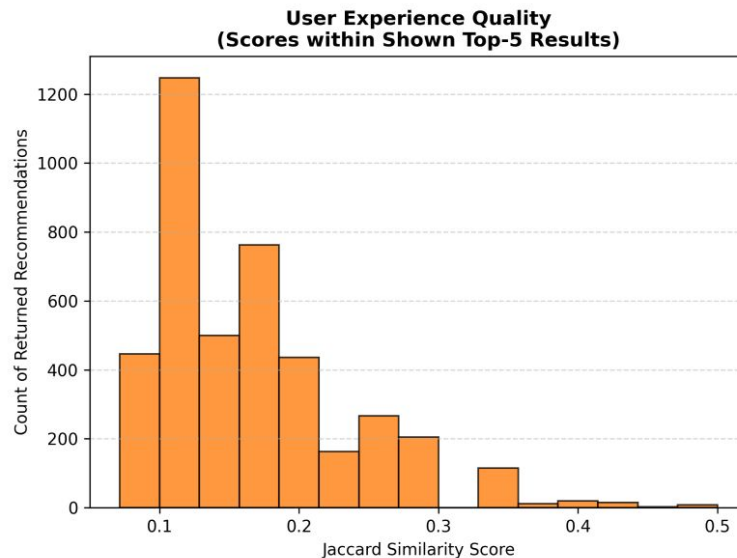
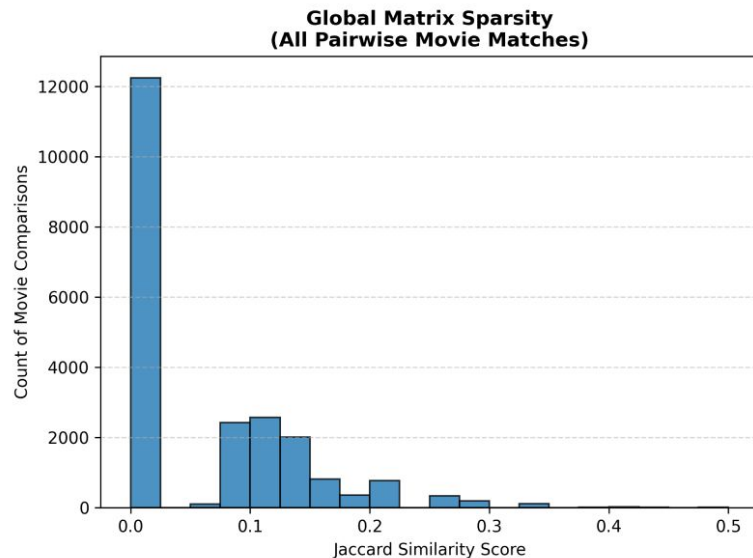
Jaccard Heuristic: similarity score based on overlaps

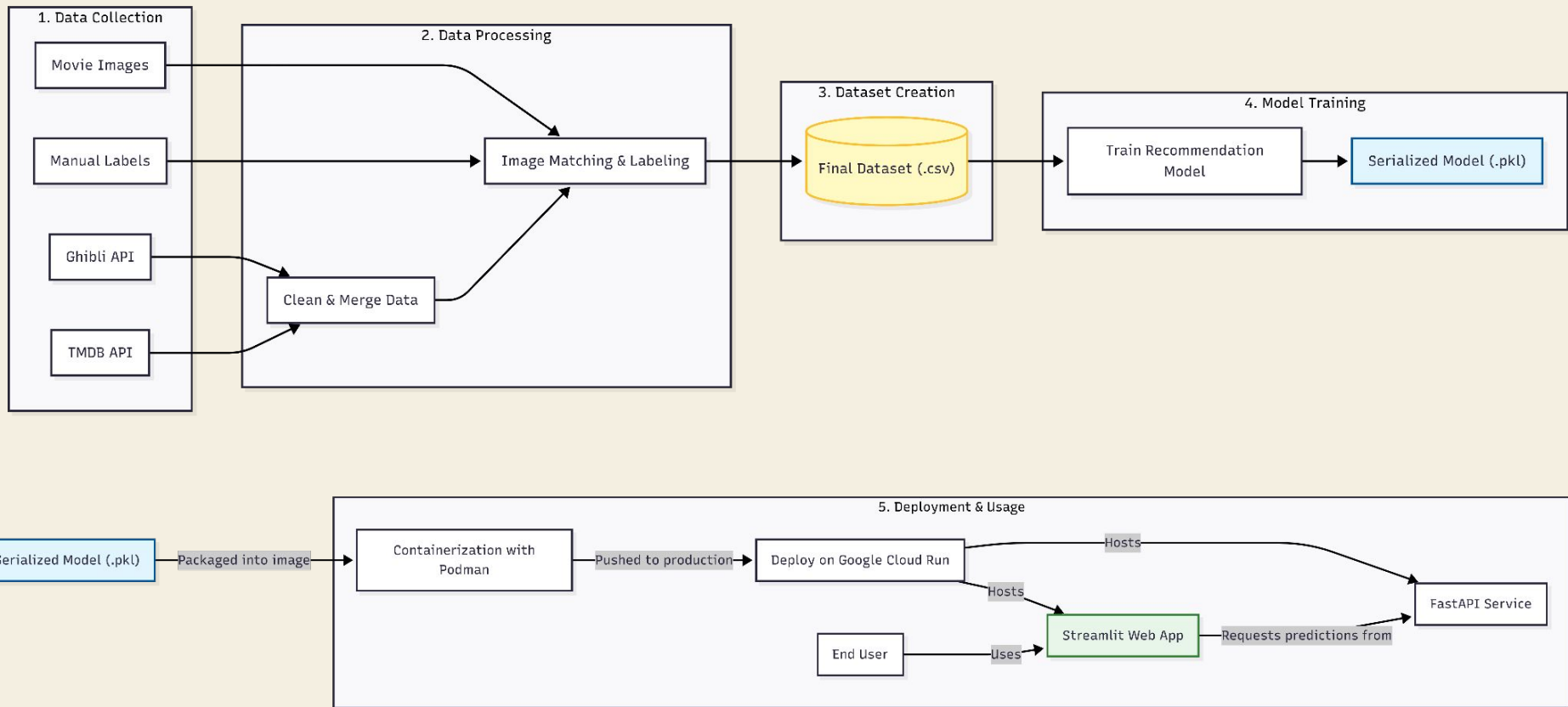
Output: Movies with highest score, sorted

Hit Rate: 91%

Mean reciprocal rank: 0.6402

Catalog Coverage: 100%





Studio Ghibli Film Explorer

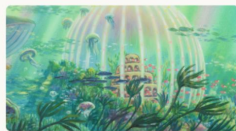
☐ Slideshow View ☒ Grid View



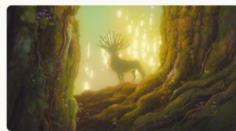
Castle in the Sky



Porco Rosso



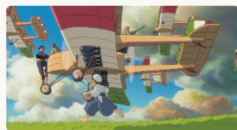
Ponyo



Princess Mononoke



Spirited Away



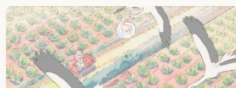
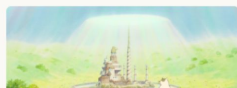
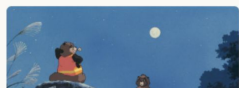
The Wind Rises



The Tale of the Princess Kaguya



Only Yesterday



Choose Your Interests

Genres

Action x



Labels

coming of age x



Species

Cat x



Recommended Films

Kiki's Delivery Service

A young witch, on her mandatory year of independent life, finds fitting into a new community difficult while she supports herself by running an air courier service.

★ 7.833



[Link to Demo](#)

AI Usage:

Google Gemini, ChatGPT – helpful for initial framework; lots of manual edits + debugging – edit file paths for successful deployment

Modeling with a Small Dataset:

Traditional ML (Random Forest) → unsupervised Jaccard Similarity heuristic

UI/UX Optimization:

Grid and Slideshow views, precise CSS for accessible + fun design

Improvements:

Add newer Ghibli films and other films directed by Hayao Miyazaki, add movie OST info